There are total 4 parts (related research and gap analysis)
1) EEG and Biomarker related
2) Binural beats
3) Wearable
4) Feature Extraction related

Related Research

1) **EEG and Biomaker related**:
   For the past several decades, researchers have studied the role of EEG as a biomarker for migraine disease. Initial studies demonstrated that delta/theta activity was higher during migraine attacks and beta was higher between attacks. Newer studies that have used the quantitative EEG method confirm evidence of cortical hyperexcitability among people who suffer from migraines, including decreased alpha activity, high-frequency oscillations, and interhemispheric coherence abnormalities \cite{VanDenHoek2024}. MEG Studies localized the areas of cortical hyperexcitability to occipital, parietal, and prefrontal cortices, which are related to sensory perception and pain processing. Notably, investigators determined that quantitative EEG occurred up to 36 hours pre-migraine and could thus be used as a prodromal biomarker \cite{Kim2023}. The study conducted by Joshi et al. \cite{JoshiMatharu2025} used EEG to create spiking and convolutional spiking neural networks to detect stress with an emphasis on developing temporal models that would provide accurate predictions while minimizing energy consumption; their convolutional spiking neural network achieved 98.61\% accuracy, 100\% sensitivity, and 98.60\% F1-score, demonstrating rapid processing speed and compactness for use with wearable devices. Van den Hoek et al. \cite{VanDenHoek2024} offered an overview of spontaneous EEG, evoked potentials, and treatment-related alterations throughout the migraine cycle to assess EEG as a marker of susceptibility, wherein preictal alterations in theta/alpha power, slowing, and connectivity were found to be consistently present, but no one secure marker was discovered; it was determined that future duties will include using home-based wearable EEG devices and implementing conventional protocols to study biomarker susceptibility to migraine. Zhang and colleagues evaluated EEG use in migraine (i.e., resting and stimulus-evoked spectral characteristics; connectivity; microstates; ML/DL) -- reporting previously seen changes (i.e., greater alpha slow; elevated delta/theta; modified networks; and the presence of useful ML classifiers) and logistical challenges (i.e., individual variance; small participant numbers) when defining reliable biomarkers. Analyzing EEG data, Bie et al. \cite{FrimpongManson2024} used both prospective and retrospective data from large sized cohorts to identify objective migraine signatures and described interictal (period of no migraine) increased occipital 12-13 Hz power as reproducible headache biomarkers. They demonstrated their KNN and SVM machine learning algorithms classified migraines and controls with 87.2\% accuracy supporting the use of occipital alpha as a headache biomarker candidate. Quartetti et al. \cite{Quartetti2025} conducted a systematic review of 36 studies identifying lifestyle factors incident to episodic migraines to develop forecasting algorithms. They concluded that ketogenic diets, supplements, and physical activity are evidence-based strategies to reduce the number of attacks and could be used to generate predictive models; however, no EEG headache biomarkers or machine learning models have been directly tested. Additionally, Zebhauser et al. \cite{Zebhauser2024}, in their meta-analysis of 27 studies using EEG/MEG techniques of individuals without migraine limits,

found a candidate EEG signature for migraine based upon increased theta power, reduced alpha to beta connectivity between brain regions to measure presymptomatic migraine; and noted these findings had significant methodological limitations including: presence of comorbidities, small sample sizes, and differential methodologies. Abdulhussein et al. Findings on SSVEPs are consistent with those of Smith et al. \cite{AbdulhusseinAldeen2024}, which suggest that rhythmic stimulation can act as a pacemaker for thalamocortical dysrhythmias in migraine. In their systematic review, Gomez-Pilar et al. \cite{GomezPilar2022} also found that EEG, MRI and/or PET imaging studies provide evidence of different brain network activity patterns between individuals with chronic and episodic migraine. Consistent differences in fast EEG/MEG bands and in networks involved in pain processing and emotional responses were identified, suggesting that functional multimodal biomarkers may provide objective markers for determining whether an individual is at higher risk for developing chronic migraine. Riederer et al. \cite{Riederer2023} published a narrative review of the EEG and MRI characteristics of migraine aura, in order to assist neurologists in diagnosis of migraines with aura versus other types of headache: typically, uncomplicated visual auras (e.g., scotomas) show normal scalp EEG, whereas complex auras (e.g., olfacto-gustatory hallucinations) typically show focal slowing or suppression of the acute phase EEG, and review various MRI "index veins," perfusion oligemia, and reversible lesions or morphology (ML) as possible differentiators between migraine aura and stroke or epileptic seizures. Göker \cite{Goker2023} proposed a novel automated migraine detection system using Welch's method (1-49Hz) and submitted that a 128 channel BiLSTM achieved an overall accuracy of 95.99\%, with a sensitivity of 0.96, a specificity of 0.95, and an F1 score of 0.96, resulting in significantly better performance than RF, SVM and LDA classifiers trained on the same EEG dataset. Kedgan et al. A systematic review and pooled prevalence meta-analysis of EEG patterns was conducted by \cite{Kedgan2025} at the different phases of aura, headache, post-headache, and inter-ictal for hemiplegic migraine. They found low amplitude, contralateral background activity during the aura phase, followed by focal slowing of the EEG during the headache phase. There was evidence of largely normal EEG during the interictal phase along with evidence of genotype-related differences. Their work provides phase-specific electrophysiological evidence for differentiating between hemiplegic migraine and stroke/epilepsy. A scoping review on EEG for diagnosing and classifying migraines was presented by Raghuraman and Joshi \cite{RaghuramanJoshi2024}, who found evidence of cortical hyperexcitability, deficits in habituation, abnormal microstates (decreased class C microstates, increased class B/D microstates), and automated AR-Burg models (up to 88.7\% accuracy); however, they concluded that methodological heterogeneity and a current focus on migraine without aura prevents the establishment of a reliable automated etiology-based diagnostic system for migraine. An evaluation of QEEG as a potential biomarker of migraine was performed by Kim et al. \cite{Kim2023}, who evaluated 720 subjects using z-score maps and found that individuals with migraines demonstrated an elevated level of overall absolute power compared to the individuals without migraines, with particularly high levels of absolute power in the beta frequency range (25-30 Hz) and these high beta frequencies predicts diagnosis and correlates with improvement, supporting the role of QEEG for migraine headaches: "Migraine patients had elevated absolute power across all.

2) **Binaural Beats**:

Melnichuk et al. \cite{Melnichuk2025} employed a sustained attention CPT with parametric variations of beta (16 Hz) and gamma (40 Hz) binaural beats. EEG entrainment was reliable, especially for gamma binaural beats, and a small improvement in hit rate was seen with optimal conditions (40 Hz, 340 Hz carrier, white noise, task-onset). However, vigilance decrement still occurs. Yang et al. \cite{Yang2025} employed a randomized crossover study of 6, 10, and 25 Hz binaural beats in college students. All binaural beats reduced heart rate/blood pressure, while 25 Hz had the strongest parasympathetic shift (nLF↓, nHF↑) but had no EEG measurement. Xiong et al. \cite{Xiong2025} performed a meta-analysis of 15 RCTs of binaural beats in the perioperative period. Anxiety reduction was substantial (SMD -1.0 to -1.4) along with moderate pain relief and hemodynamic effects compared to silence/non-BB audio. However, evidence certainty is moderate.Kahathuduwa et al. \cite{KahathuduwaBlume2025} created real-time EEG-guided binaural beats and showed an enhancement of relaxation/cognition compared to controls. They also showed the validity of adaptive entrainment of binaural beats. Shamsi et al. \cite{Shamsi2024} performed a systematic review of binaural beats in pain management. Alpha/delta-alpha binaural beats are effective in acute procedural pain but show inconsistent results in chronic pain,with low evidence quality because of bias and heterogeneity. By using wearable EEG to compare stationary binaural beats, panning sounds, and beeps, Sudre et al. \cite{Sudre2024} demonstrated that spatial motion produced stronger relaxation/EEG effects than beat illusion alone.Ingendoh et al. \cite{Ingendoh2023} examined 14 EEG studies and concluded that only 5 of them demonstrated entrainment (none of them convincingly in beta), recommending that standardized procedures be used before making generalizations. In their study of 0.25/1 Hz beats during naps, Fan et al. \cite{Fan2024} found that 0.25 Hz shortens N2/N3 onset without strong entrainment, indicating behavioral slow-wave facilitation. Beats were tested during dissection by Mosheshe et al. \cite{Mosheshe2024}, who found that they helped BBs cope with acute stress in the absence of an EEG and decreased student anxiety in comparison to controls. Longer beat sessions significantly reduced HR/LF-HF and improved mood via HRV/BRUMS, suggesting dose-response relaxation effects, according to Chockboondee et al. \cite{ChockboondeeJatupornpoonsub2023}. Dabiri et al. \cite{Dabiri2022} demonstrated that delta beats enhanced the mood and quality of sleep of pilot subjects, supporting BB sleep aids pending EEG validation. According to Corona-González et al. \cite{CoronaGonzalez2021}, customized theta/beta beats increased target power more than fixed via EEG, favoring individual tuning.Al-Shargie et al. \cite{AlShargie2022} used fNIRS/16 Hz beats in Stroop tasks to decrease stress (44\% less alpha amylase) and increase accuracy and speed through prefrontal networks. 40 Hz beats increased gamma/alpha and reduced delta activity, improving working memory compared to controls, according to Wang et al. \cite{WangZhang2022}. A meta-analysis of BBs in cognition by Basu and Banerjee \cite{BasuBanerjee2023} showed small improvements in memory and attention, although results are suspect due to bias, heterogeneity, and lack of evidence of entrainment. Alpha beats specifically activated EEG/cognition compared to white noise, according to Hamid et al. \cite{Hamid2025}.

3) **Wearable:**

Petru\v{s}i\'{c} et al. \cite{Petrusic2025} have characterized EEG-based brain--computer interface (BCI) technology as an ``emerging and scalable solution for brain monitoring, rehabilitation, and assistive technologies," with the significant benefit of ``enabling real-world neural signal acquisition without the associated risks of invasive systems." They have also categorized wearable EEG systems as scalp-EEG, forehead-EEG, and ear-EEG systems, with scalp-EEG systems offering higher cortical resolution and motor imagery/assistive control classification accuracy of 90--96\%. However, they also note key limitations, including susceptibility to artifacts, inter-subject variability, electrode impedance instability, and the absence of standardized benchmarking frameworks \cite{Petrusic2025}.

He et al. \cite{He2023} show that commercial wireless EEG systems remain highly heterogeneous with respect to EEG channel count, sampling rate, battery life, and price, ranging from \$240 to \$30{,}000. While this diversity reflects rapid innovation in hardware, He et al. also emphasize the lack of standardization in signal validation methods and real-world artifact handling, which can constrain reliable application of these systems \cite{He2023}. In parallel, Muhammad et al. \cite{Muhammad2023} introduce a cost-effective OpenBCI-based headset with a 16-channel configuration, achieving 82.9\% performance with a rapid setup time of 2 ± 0.5 minutes at an estimated cost of \$299. A reduced 3-channel version achieves comparable performance of 82.0\%. Nevertheless, the evidence base for wearable EEG systems remains limited by small-scale validation and evaluation, particularly with respect to performance consistency across heterogeneous users \cite{Muhammad2023}.

Danelakis et al. \cite{Petrusic2025} extend the wearable concept beyond EEG devices through digital phenotyping, which integrates wearable technology, smartphone-based monitoring, and AI-based analytics to monitor neurological disorders such as migraine under free-living conditions. The forecasting performance reported by Danelakis et al. ranges from 62\% to 84\%, and the associated behavioral signatures of ictal states are supported by observational findings showing significant reductions in physical activity during migraine episodes, with statistical significance ranging from $p < 0.01$ to $p < 0.001$ \cite{Petrusic2025,VanDerDonckt2025}. Furthermore, Danelakis et al. \cite{DanelakisStubberud2025} report that AI-supported diagnostic models in headache studies demonstrate accuracies between 62\% and 99.7\%, while also highlighting practical limitations including small sample sizes, overfitting risk, wearable signal noise, limited interpretability, and inconsistent validation procedures \cite{DanelakisStubberud2025}.

Taken together, these findings suggest strong potential for integrating wearable EEG platforms and digital phenotyping with AI-based analysis for cost-effective and personalized neurological monitoring. However, successful clinical translation will likely require larger multicenter datasets, standardized benchmarking and validation frameworks, robust signal quality evaluation methods, and interpretable machine learning models \cite{Petrusic2025,DanelakisStubberud2025,He2023}.

4) **Feature Extraction related:**

Yuvaraj et al.\cite{Yuvaraj2023} investigated the performance of statistical, wavelet, fractal dimension, Hjorth and higher-order statistics features in the recognition of emotions in EEG data. Their analysis showed that the combination of fractal dimension and CART algorithm provided a high classification accuracy (≈ 85\%) for valence/arousal; thus, allowing for the creation of an EEG-based emotion recognition algorithm suitable for classification in real-time. Kalashami et al. \cite{Kalashami2022} extracted 344 handcrafted EEG features (frequency band powers, asymmetry, time domain) from the DEAP database and employed conditional Wasserstein Generative Adversarial Networks (CWGAN) to augment their dataset with 5000 new samples. The deep neural network was able to classify valence with a classification accuracy of 67.4\% and arousal with an accuracy of 71.9\%. Luo et al. \cite{Luo2024} reconstructed 8-level wavelet packets (db4) from EEG data collected during virtual reality-induced motion sickness, and used these feature sets as input to a gated recurrent unit (GRU) model. The resulting classification accuracy was 84.9\%, and achieved a Pearson linear correlation coefficient (PLCC) of approximately 0.90 and a Spearman rank-order correlation coefficient (SROCC) of approximately 0.92 for regression.Rakhmatulin et al. \cite{Rakhmatulin2024}: explored CNN architectures (EEGNet, cascaded TS-SEFFNet) on EEG extraction with ICA/PCA/CSP preprocessing achieving greater than 90\% motor imagery / P300 task. Chaddad et al. \cite{Chaddad2023} from assessment of EEG pipelines(denoising,time-frequency,HOS,nonlinear features), that were processed showed SVM/CNNs achieved over 90–99\% accuracy across epilepsy, depression, and sleep with appropriate extraction. Singh and Krishnan \cite{Singh2023} reviewed trends in EEG feature datasets, noting that top-performing spatial extractors were variants of CSP methods with an accuracy range of 80–95\% BCI versus a time/frequency only results give 70–80\%.

Gap Analysis:

1)EEG and Biomarkers and beats gaps:

EEG methods show real promise as biomarkers for migraines and for brainwave entrainment, but there are still some big gaps holding things back from actual clinical use. Take Joshi et al., for example. They based their work on the PhysioNet database, which just doesn't cover enough diverse populations. Plus, they used new neuromorphic hardware that isn't ready for the clinic yet. These two factors really get in the way of putting their findings into practice. Then there's the study by van den Hoek et al.As a narrative review, \cite{VanDenHoek2024} is devoid of quantitative meta-analysis, gamma band exploration, empirical data, wearable evaluations, and a suggested standardization protocol.

Zhang et al. \cite{Zhang2023} and Raghuraman and Joshi \cite{RaghuramanJoshi2024} both ran into trouble with inconsistent EEG methods, small sample sizes, and visual bias—all of which make it tough for others to repeat their work or agree on reliable biomarkers. Bie et al. \cite{FrimpongManson2024} had issues too: their cohort was small, participants were still on medication, the timing between attacks wasn't precise, and they didn't separate migraine subtypes, so it's hard to apply their findings more broadly.

Quartetti et al. \cite{Quartetti2025} faced a mix of problems—different study designs, confounding factors like COVID they didn't account for, and no integration of EEG or detailed machine learning datasets. All of that gets in the way of building solid forecasting models. Zebhauser et al. \cite{Zebhauser2024} didn't control well for comorbidities, their studies were underpowered (average sample size was just 62), their methods varied a lot, and they didn't use blinding. With all this, it's tough to pin down migraine-specific M/EEG changes.

Abdulhussein et al. \cite{AbdulhusseinAldeen2024} run into trouble with volume conduction artifacts—they make it hard to know if deep brain structures are actually being modulated or if it just looks that way. They need more than just detection of dysfunction; they have to show real therapeutic changes. Gomez-Pilar et al. \cite{GomezPilar2022} face a bunch of obstacles too: their small sample sizes, changing methods, and cross-sectional approach make it tough to say anything solid about what causes migraine chronification. Plus, they don't have clear topographical hypotheses or use EEG and fMRI together, which would really help.

Riederer et al. \cite{Riederer2023} struggle to capture short auras as they happen. Since they mostly study tricky or unusual cases, it's hard to apply their findings to the average migraine patient. Goker \cite{Goker2023} deals with lots of high-dimensional data, so BiLSTM models work better, but old-school machine learning just can't keep up. Their use of holdout splits also risks overfitting.

Kedgan et al. \cite{Kedgan2025} mostly rely on case reports and series, which come with high bias and weak evidence—so their results don't really apply to migraines outside hemiplegic cases. Kim et al. \cite{Kim2023} only use data from one center, and they don't fully account for medication effects or whether people are in the middle of a migraine or not, making it hard to trust QEEG as a diagnostic tool on its own.

As for brainwave entrainment and binaural beats \cite{Melnichuk2025,Yang2025,Xiong2025,KahathuduwaBlume2025,Shamsi2024,Sudre2024,Ingendoh2023,Fan2024,Mosheshe2024,ChockboondeeJatupornpoonsub2023,Dabiri2022,CoronaGonzalez2021,AlShargie2022,WangZhang2022,BasuBanerjee2023,Hamid2025}, these studies all run into similar problems. There's no consistent evidence that entrainment actually happens, protocols are all over the place (different frequencies, durations, and whether noise is mixed in), and they usually use small groups of young adults. Controls and blinding are often weak, and most only rely on self-reports without real EEG validation. Without standardized randomized trials or personalized closed-loop methods, you just can't generalize these results to treating clinical pain like migraine.

2)Wearable:

Despite the rapid advancements made in wearable EEG technology and digital phenotyping methodologies, from the existing body of literature, there are significant gaps identified in the methodologies. Petru\v{s}i\'{c} \cite{Petrusic2025} has emphasized that there is a significant gap that needs to be filled in the existing methodologies, and despite the tremendous potential of digital phenotyping technology for the continuous monitoring and prediction of migraines, it still suffers from heterogeneous methodologies, non-diverse sample populations, and no external validation. The absence of standardized methodologies for data acquisition, processing, and validation of multimodal data is one of the significant gaps identified in the existing methodologies. Similarly, Danelakis et al. \cite{DanelakisStubberud2025} have identified that despite the high performance of AI-based diagnostic systems, they still suffer from non-validation and overfitting due to non-diverse sample populations.

From the hardware perspective, He \cite{He2023} identified that there are significant gaps identified in the existing methodologies of wearable EEG technology. The existing methodologies of wearable EEG technology suffer from heterogeneous channel configuration, sampling rate, battery capacity, and quality assessment methodologies. The absence of standardized benchmarking methodologies for the assessment of artifacts and signal quality is one of the significant gaps identified in the existing methodologies. However, the lack of standard methods for benchmarking artifacts, especially when it comes to handling and evaluation of fidelity signals, poses a question of reliability, especially when it comes to consumer-grade solutions for such handling and evaluation scenarios \cite{He2023}. In the case of the use of wearable devices for the handling and evaluation of behavioral aspects, Van Der Donckt, in his work, presents the possibilities of using wearable devices like actigraphy for the objective identification of significant reductions in physical activities that occur during the ictal migraine phases. However, the results are based only on small-scale experiments, without any comparison to healthy subjects or predictive modeling for early intervention/detection \cite{VanDerDonckt2025}. In the case of the use of digital phenotyping for the handling and evaluation of migraines, as presented by Danelakis et al. \cite{Petrusic2025}, it is clear that digital phenotyping is the way forward for the handling and evaluation of migraines, which is a paradigm shift in the handling and evaluation of migraines. However, the fact that most of the applications of digital phenotyping, as presented, remain conceptual is also acknowledged.

Taken together, the literature suggests that there are four key gaps: (1) the lack of standardized signal validation and benchmarking tools for wearable EEG technology, (2) the lack of large-scale, multi-center clinical data sets that include multimodal physiological and behavioral signals, (3) the lack of development of interpretable and externally validated AI models for migraine prediction and treatment, and (4) the lack of closed-loop systems that combine continuous monitoring and adaptive therapeutic approaches. Closing these gaps will require a focus on methodological standardization, real-world validation, multimodal wearable data integration, and the development of transparent and trustworthy machine learning architectures.

4)Feature Extraction related:

Even with all the progress in EEG feature extraction, big problems stick around. Yuvaraj et al. \cite{Yuvaraj2023} ran into trouble with subject-dependent validation and skipped deep learning tests altogether, so their results don't really generalize. Kalashami et al. \cite{Kalashami2022} had issues with fuzzy labels and their GAN models turned out to be pretty sensitive. Luo et al. \cite{Luo2024} worked with tiny samples, and their results got skewed by subjective SSQ bias. Rakhmatulin et al. \cite{Rakhmatulin2024} struggled to handle dataset differences and couldn't get things running in real time. Chaddad et al. \cite{Chaddad2023} didn't really solve the core noise and variability headaches. And Singh and Krishnan \cite{Singh2023} just pointed out the obvious accuracy is still lacking, and there's no real standard for any of this.

Summary:

To guide the theoretical and technical underpinnings for our proposed EEG-based intelligent migraine management system we conducted a systematic review of four significant areas of research: (1) EEG as a migraine biomarker, (2) Brainwave entrainment and binaural beats, (3) Wearable devices, (4) Feature extraction methods and approaches. The presented domains were chosen carefully because they cover the key elements of our project—biomarker discovery, signal processing, intelligent modeling and real-world implementation.

First, we cover the literature on EEG as a migraine biomarker that details how EEG signals can be used to identify cortical hyperexcitability and other neural patterns associated with migraines. Knowledge of these neurological signatures is key to rationalizing EEG as the primary source of data within our system.

Second the studies on brainwave entrainment and binaural beats, lend perspective to how specific frequency stimulations can affect our brain. Rather than focusing solely on the specific data we plan to acquire, our project also considers potential intervention strategies so a review of this domain can inform us as to how controlled auditory stimulation may affect the migraine brainwave abnormalities we hope to observe.

Third, studies of wearable devices could inform the feasibility of continuous, non-invasive EEG monitoring in naturalistic environments. Given that our system targets practical, portable implementation, knowledge of hardware constraints (e.g. sensor placement, signal quality) and user comfort is paramount for the development of a performant hybrid interaction paradigm.

Fourth, it is scientifically needed to revise the best approaches for extracting features because model performance directly depends on meaningful features (the time domain and frequency-domain as well as the time–frequency features). The key is using proper feature generation to extract the relevant signal characteristics from a migraine perspective.

These four domains were systematically chosen, however, due to how closely they aligned with the scientific, technical and practical aspirations for our project. Combined, they enable an integrative conception of neural implementational substrates, signal processing techniques, intelligent modeling frameworks and real-world implementation approaches. By carefully studying these domains we ensure that the framework we propose is scientifically justified, technically robust and practically feasible.

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